Table of Contents

1 Introduction: Computing Before Computing 3

1.1 An Observation Spanning Several Decades 3

1.2 The Auditory System as a Multi-stage Analogue Processor 3

1.3 Animals as Extreme Cases of the Principle 5

1.4 The Structural Cause of the Discrepancy 5

1.5 Parallels to Hybrid Photonics 6

1.6 Structure of this Document 6

1.7 ξ is Encoded in Physics: Crystals, Cochlea, and Natural Signals 8

2 Does the Heisenberg Uncertainty Principle Apply to Hearing? 10

2.1 The Uncertainty Relation and its Preconditions 10

2.2 The Cochlea as an Active Non-linear System 10

2.3 Phase-Locking: Frequency Information from Time 11

2.4 The Auditory System as Quantum Computer — a Precise Analogy 11

2.5 Design Principles for Technical Hearing Systems 12

2 Conclusion: The Trained Ear as Superior AI Detection 14

2.1 Why Technical Detection is Structurally Disadvantaged 14

2.2 What a Trained Ear Detects 14

2.3 Training Instead of Trust in Technical Solutions 15

2.4 The Trained Ear as Superior Detector — for Decades 15

Document 170

Analogue Pre-processing as a Computational Principle

The Auditory System, the Cochlea, and Hybrid Photonic Systems

Johann Pascher · March 2026

Core thesis: The persistent superiority of the biological auditory system over digital systems in certain tasks (cocktail-party problem, transient events, source separation) has a structural cause: the auditory system uses physical relaxation as a computational principle. The cochlea performs spectral decomposition because its mechanics are so constructed — not because something is calculated. The same principle underlies Coherent Ising Machines and hybrid photonics. This opens a new design space for audio processing systems.

1 Introduction: Computing Before Computing

1.1 An Observation Spanning Several Decades

Anyone who has worked with digital audio signal analysis over several decades inevitably encounters a persistent discrepancy: the ability of a human — or an animal — to distinguish sounds in fractions of a second, to classify them, to isolate them from overlapping signals and trace them to their source, is not fully matched by digital systems despite enormous advances in computing power and data volume.

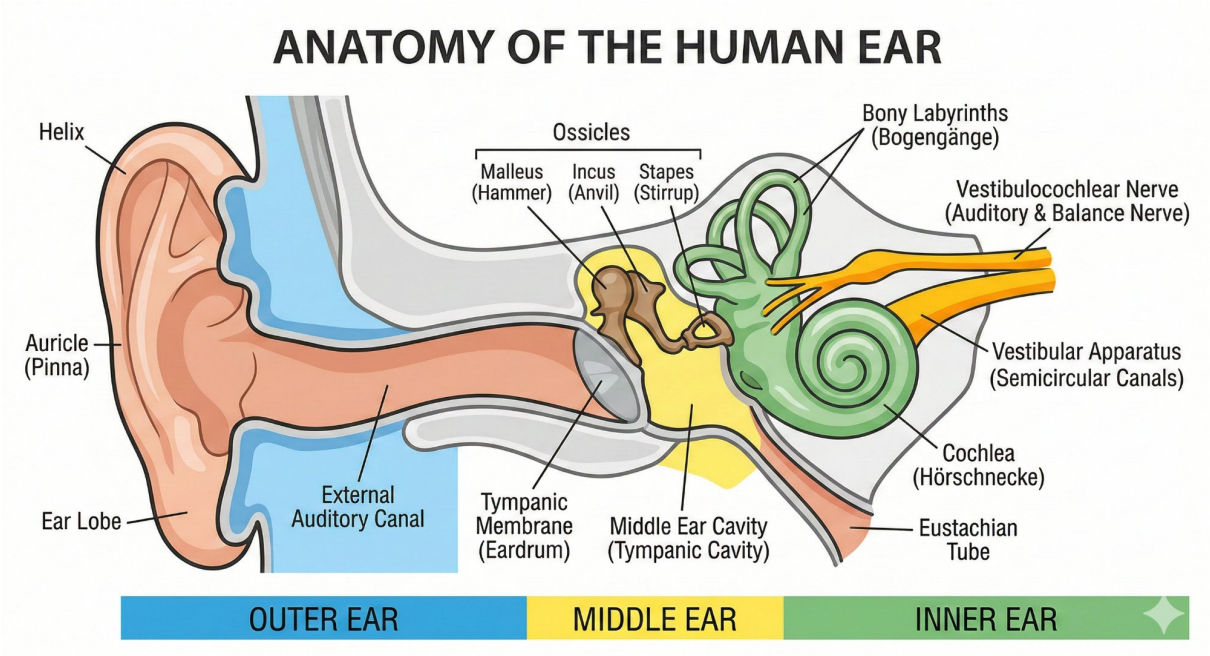
Speech recognition has improved dramatically in recent years. Nevertheless, there are domains in which the human or animal auditory system remains superior:

- Separating simultaneous speakers in a loud, diffuse sound field (cocktail-party problem)
- Identifying a single instrument within a full orchestral texture
- Detecting short transient events below the frame resolution of digital systems
- Hearing well below signal-to-noise ratios at which digital methods fail
- Immediate source localisation in three-dimensional space from two ear signals

This work argues that the cause of this discrepancy is structural — not a matter of better algorithms, larger training sets, or faster processors. The biological auditory system follows a fundamentally different processing principle: physical relaxation as a computational principle.

1.2 The Auditory System as a Multi-stage Analogue Processor

The auditory system is not a biological microphone followed by a neural computer. It is a multi-stage analogue processing system that substantially structures the signal before a single nerve cell fires. Each stage uses physics as a pre-processor.



Anatomy of the human ear: helix, ear canal, tympanic membrane, ossicles (malleus, incus, stapes), cochlea, and vestibulocochlear nerve. Every stage is an analogue pre-processor.

Pinna and External Ear Canal

The geometry of the pinna filters incoming sound waves as a function of frequency and direction — passively, without energy expenditure. This geometric filtering encodes spatial information (elevation angle, front-back discrimination) into the amplitude spectrum. A digital system seeking the same information must compute it through cross-correlation of microphone signals. The pinna computes it through physics.

Middle Ear

The middle ear performs impedance matching between air and perilymph and actively compresses the dynamic range via the stapedius reflex. This non-linear compression — fast for transient events, slow for continuous signals — corresponds to an adaptive analogue compander responding to signal statistics in real time. No digital algorithm can replicate this function without latency.

Cochlea: Mechanical Fourier Analyser

The basilar membrane of the cochlea is a spatially distributed resonator. Each point along its length resonates at a different frequency: approximately 20 kHz at the base, 20 Hz at the apex, continuously and logarithmically graded. A sound wave is decomposed onto this spectral analysis system without any computation step — the Fourier decomposition happens because the mechanics of the system are so constructed.

This is fundamentally different from the digital Fast Fourier Transform: the FFT is an algorithm operating sequentially on stored sample values, requiring a minimum window length. The cochlea analyses continuously, in real time, without windows, without a clock.

Outer Hair Cells: Active Feedback

The decisive feature that drives cochlear resolution far above the passive mechanical limit is the outer hair cells. They are electromechanically active: the membrane protein prestin contracts and expands as a function of electrical potential, selectively amplifying the mechanical vibration of the basilar membrane through positive feedback. The auditory system contains a biological analogue feedback amplifier with frequency-selective gain — on every frequency channel simultaneously.

The result: the signal that reaches the inner hair cells and is converted into neural impulses is already spectrally analysed, logarithmically compressed, spatially encoded, and non-linearly filtered. The processing time is less than 10 milliseconds. The brain receives no raw data — it receives an already structured representation of the acoustic scene.

1.3 Animals as Extreme Cases of the Principle

In certain animal species the principle becomes visible in its purest form. Bats solve spatial reconstruction problems via echolocation in milliseconds that a digital system based on the computing power and brain size of these animals could never solve. Dolphins analyse ultrasonic reflections so precisely that they can locate objects behind obstacles and distinguish materials. The nervous systems of these animals are not large enough for this — if one assumes that computing means the same as in a digital computer.

> *Physics computes. The nervous system interprets.*

1.4 The Structural Cause of the Discrepancy

The persistent superiority of the biological auditory system in certain tasks is not a training or data problem. It has a structural cause that can be formulated on three levels:

Level 1: Pre-processing through physics. The biological system solves an essential part of the processing problem through the physical construction of the sensor — not through algorithms. Digital audio technology separates sensor (as linear as possible) and processing (everything follows numerically afterwards). This separation wastes the computational potential of the physical system.

Level 2: Temporal continuity. Digital systems work with sampled, stored, then processed frames. Between capture and analysis there is always a window latency. The biological system operates continuously: the basilar membrane responds to signals shorter than any sensible digital frame.

Level 3: Scaled complexity. The cocktail-party problem remains demanding for digital systems. The human auditory system solves it effortlessly because source separation already begins in the physical pre-processing: spatial filtering by the pinna, interaural time differences, harmonic structure of voices. The digital system must reconstruct this physical information after the fact.

1.5 Parallels to Hybrid Photonics

The hybrid photonic computing systems that have emerged in recent years — in particular the Coherent Ising Machine (CIM) — pursue the same fundamental principle on a different physical substrate.

In a CIM, combinatorial optimisation problems are encoded as Ising Hamiltonians. Optical parametric oscillators adopt phase states (0° or 180°) corresponding to the spin states ±1 of the Ising model. The system relaxes physically to an energy minimum — this is the solution to the optimisation problem. A digital computer then reads and verifies the result.

Cochlea / Auditory system	CIM / Hybrid photonics
Mechanical resonance of the basilar membrane	Optical resonators, parametric oscillators
Frequency separation through geometry	Problem encoding through phase coupling
Analogue, massively parallel, no clock	Analogue, massively parallel, speed of light
Outer hair cells: active feedback	Phase-sensitive amplifier, feedback loop
Hair cells: analogue → neural spike	Measurement: analogue → binary (0°/180°)
Brain processes pre-structured information	Digital computer reads relaxed state

Structural analogy between the biological auditory system and CIM.

> The common principle: A physical system with suitable internal structure relaxes into a state that solves a hard analytical or combinatorial problem. The processing is the physics — not a computation that runs afterwards on physical measurement data. The cochlea solves continuous spectral decomposition through mechanical resonance. The CIM solves NP-hard Ising problems through optical phase relaxation. Both are hybrid systems: the analogue substrate handles the computationally expensive part; a digital system handles encoding, decoding, and sequential logic.

1.6 Structure of this Document

This document develops these analogies in several directions and draws out what follows for the design of future audio processing systems.

Chapter 2: The Cochlea as an Information System. Mechanical properties, active amplification, logarithmic scaling, temporal resolution.

Chapter 3: Structural Limits of Digital Audio Analysis. Where and why digital systems encounter limits that cannot be overcome by more computing power.

Chapter 4: Hybrid Photonic Computing Systems. Coherent Ising Machines, neural photonic networks and their structural kinship with the biological auditory system.

Chapter 5: Outlook — Architectures for Analogue Audio Cognition. Analogue pre-processing sensors, neuromorphic cochlea chips, photonic-neural hybrid systems for real-time audio cognition.

1.7 ξ is Encoded in Physics: Crystals, Cochlea, and Natural Signals

The preceding sections described the cochlea as a physical pre-processor. There is a deeper level: The relevant mathematics is not encoded in algorithms — it is encoded in physics itself. This applies to crystals as much as to the basilar membrane, and in T0 theory both share the same origin: the geometric parameter ξ.

Crystals: The Geometry is the Solution

A crystal does not solve an eigenvalue problem — it is the solution. Band structure, phonons, and the dielectric function arise because the periodic lattice decomposes waves into eigenmodes according to its geometry. No algorithm computes — physics propagates.

The lattice constant is determined by atomic radii and bond lengths at the atomic length scale a_0. In T0 this scale is the rung ξ^7 of the ξ-power ladder:

$$a_0 \approx \xi^{-6,326} \cdot \ell_P \quad R_\infty \approx \xi^{6,956} \cdot E_P$$

Quantity	Value	ξ-Exponent	Meaning
Bohr-Radius a_0	0.0529 nm	$\xi^{-6,326}$	Atomskala / atomic scale
NaCl-Gitterkonstante	0.282 nm	$\xi^{-6,514}$	Ionenkristall / ionic crystal
Si-Gitterkonstante	0.543 nm	$\xi^{-6,587}$	Halbleiter / semiconductor
Rydberg R_∞	**13.598 eV**	**$\xi^{+6,956}$**	**Atomenergie / Cochlea-Skala**

All atomic length scales and lattice constants lie on rung $|\xi^7|$ of the ξ-ladder.

The Cochlea as a Biological ξ^7 Resonator

The basilar membrane consists of collagen, proteoglycans, and water. The elastic modulus of this tissue arises from chemical bonds between molecules — i.e. the molecular scale, the ξ^7 rung. The stiffness at position x along the cochlea:

$$k(x) \sim E \cdot t(x)^3 / w(x) \Rightarrow f(x) \propto \sqrt{k(x)/\rho}$$

The elastic modulus E is determined by chemical bond energies. Chemical bond energies are multiples of R_∞ (Doc. 169). R_∞ lies on rung ξ^7. The cochlea is a ξ^7 system analysing ξ^7 frequencies. The geometry is biologically optimised; the scale is set by ξ — as with crystals.

The Greenwood Map as ξ-Ladder in Biological Matter

The tonotopic map of the cochlea follows the Greenwood function $f(x) = A \cdot 10^{a \cdot x}$ with $a = 0.06\ \text{mm}^{-1}$: one octave corresponds to $\Delta x \approx 5$ mm. The full hearing range of 10 octaves corresponds on the ξ-ladder to:

$$10 \times \log_\xi(2) = 10 \times (-0,0777) \approx -0,78\ \xi\text{-Stufen}$$

The logarithmic structure of the Greenwood map is the logarithmic structure of the ξ-ladder — physically realised in biological tissue.

1/f Noise, ξ-Ladder, and Optimal Encoding

Natural sound signals have a 1/f spectrum (pink noise): equal energy per octave. This is the statistical signature of self-similar physical processes (speech, music, ambient sounds). The ξ-ladder is logarithmically equidistant: each step multiplies the energy by $\xi^{-1} = 7500$. The ξ-ladder therefore has the same structure as pink noise. The cochlea is optimally tuned for signals with ξ-ladder statistics — because it is itself a realisation of the ξ^7 rung.

Three Levels of ξ Encoding

Level	Kristall	Cochlea
Material basis	Ionic bond, $a_0 \sim \xi^{-63}$	Collagen bond, E-modulus from chem. bonds
ξ-Stufe	Lattice constant $\sim \xi^{-65}$	Membrane stiffness $\sim \xi^7$
Computing principle	Wave propagation in potential	Mechanical resonance along membrane
Result	Band structure, phonon modes	Tonotopic frequency map
What is computed	Nothing — physics propagates	Nothing — mechanics relaxes

The fundamental principle in T0 language: ξ = 4/30 000 is the geometric packing deficit of the 3D lattice from which all fundamental constants follow. Crystal lattices, Bohr radius, Rydberg energy, elastic modulus of biological tissue, cochlear frequencies — all lie on the ξ^7 rung. Every physical system at this rung computes automatically at the atomic scale — by its mere existence.

Mathematics is encoded in ξ-geometry. Physics realises this geometry in matter. Computing is propagation.

Central thesis: The next qualitative improvement in machine auditory processing will not come from larger neural networks, but from a return to the principle that nature has used for hundreds of millions of years — physics as pre-processor, digital technology as interpreter.

2 Does the Heisenberg Uncertainty Principle Apply to Hearing?

> Core statement: The acoustic time-frequency uncertainty relation $\Delta t \cdot \Delta f \geq 1/4\pi$ applies to passive linear systems. The cochlea is neither. As an active, non-linear, coherent wave system it effectively undercuts the uncertainty — structurally analogous to squeezed light in quantum optics. The auditory system does not realise quantum computing in the technical sense, but the same processing principle: coherent active wave physics without the limitations of passive classical systems.

2.1 The Uncertainty Relation and its Preconditions

The time-frequency uncertainty relation $\Delta t \cdot \Delta f \geq 1/4\pi$ is not an experimental result — it is a mathematical theorem, a direct consequence of Fourier analysis for square-integrable functions. It applies to every passive linear system that measures a wave field.

A microphone is passive and linear — it measures pressure fluctuations without intervening in the sound field. A digital FFT is a linear transformation over a finite window. For the system microphone + FFT the uncertainty applies strictly. The cochlea is neither passive nor linear.

2.2 The Cochlea as an Active Non-linear System

Energy Injection by Outer Hair Cells

The outer hair cells inject mechanical energy into the vibration system of the basilar membrane. The membrane protein prestin changes cell length depending on membrane potential, at frequencies up to 70 kHz. This is not a passive measurement process — it is an active intervention in the signal. The system is not an observer of the sound field. It modifies the sound field it analyses.

Squeezed Light as a Structural Analogy

In quantum optics, for amplitude and phase of a light field: $\Delta X_1 \cdot \Delta X_2 \geq 1/4$. Squeezed light reduces ΔX_1 below the vacuum limit by correspondingly increasing ΔX_2. This does not violate the uncertainty relation — it shows the product is a minimum, not a symmetric bound. The system can be asymmetrically optimised when actively controlled.

The outer hair cells do the same acoustically: They push the frequency uncertainty Δf below the passive limit by sharpening frequency resolution through positive feedback. The price is energy expenditure — not loss of time resolution.

Property	FFT (passiv, digital)	Cochlea (aktiv, analog)
Mechanism	Linear transform over window	Physical resonance, continuous

Property	FFT (passiv, digital)	Cochlea (aktiv, analog)
Time window	Discrete, finite, explicit	None — resonance runs continuously
Freq. resolution	$\Delta f \geq 1/T_{+}$ enster	Resonator quality Q, actively enhanced
Time resolution	$\Delta t = 1/f_s$ (sample rate)	Onset detection, independent of freq. analysis
Phase information	Lost after FFT	Retained in spike timing (phase-locking)
Uncertainty	$\Delta t \cdot \Delta f \geq 1/4\pi$ (hard)	Effectively undercut by active feedback

The auditory system undercuts the uncertainty not by breaking the mathematics — but by being a different physical system.

2.3 Phase-Locking: Frequency Information from Time

The auditory system uses two fundamentally different mechanisms in parallel:
Place coding: which nerve fibre fires encodes frequency by the location of basilar membrane excitation — broadband, limited resolution.
Temporal coding: auditory nerve fibres fire preferentially at a specific point in the vibration cycle (phase-locking, up to ~4 kHz). The brain analyses the time intervals between spikes — the periodic structure of the signal is retained in the neural code. Frequency resolution through temporal coding exceeds place coding at low frequencies by more than an order of magnitude.

> *The FFT collapses the signal into a spectrum — after which phase information is gone. The cochlea does not collapse. It retains temporal fine structure in parallel with frequency information, because both rest on different physical mechanisms — not on the same mathematical trade-off.*

2.4 The Auditory System as Quantum Computer — a Precise Analogy

Provocative thesis: A quantum computer is not located in the brain — it is located in the auditory system. This is to be understood as an analogy, not as an ontological claim about quantum states in biological matter. It is, however, structurally more precise than it initially appears.

Principle	Quantencomputer	Cochlea / Auditory system
Superposition	Qubit in superposition $\|0\rangle + \|1\rangle$	All frequency channels simultaneously active
Coherence	Phase relations between qubits preserved	Phase-locking: phase structure in spike timing

Principle	Quantencomputer	Cochlea / Auditory system
Active control	Error correction, quantum gates	Outer hair cells: freq.-selective feedback
Non-linearity	Quantum gates non-linear, measurement collapses	Analogue → spike: threshold operation
Uncertainty	Heisenberg, actively managed (squeezed states)	Acoustic uncertainty effectively undercut
Measurement	Collapse to eigenstate	Mechanics → receptor potential → spike

What the analogy means: The auditory system realises the same processing principle as a quantum computer: coherent wave physics with active control. It is not constrained by the limits of passive classical systems. A quantum computer outperforms a classical computer not because it is faster, but because it realises a different computational principle. The auditory system outperforms microphone + FFT for the same reason.

2.5 Design Principles for Technical Hearing Systems

Five concrete design principles follow from this analysis:

Principle 1: Active resonators instead of passive filters. Frequency resolution must arise from resonator quality Q, not window length. Active feedback resonators, not FIR/IIR filters.

Principle 2: Temporal coding parallel to place coding. The temporal fine structure of the signal must be preserved alongside frequency information. Spike-timing-based representations are more fundamental than spectrograms.

Principle 3: Non-linearity as a feature. Compression and non-linearity are not artefacts to be compensated. They enable simultaneous processing of large dynamic ranges.

Principle 4: Continuous processing without frames. Window-based analysis is a workaround of the digital paradigm. The target architecture is asynchronous and causal — no frame, no latency.

Principle 5: Hybrid architecture. Analogue wave processing system + digital interpretation system. The boundary lies where physics ends and symbol processing begins.

A quantum computer is not located in the brain.
It is located in the auditory system.

Not because quantum mechanics governs it,
but because the same processing principle is realised:
Coherent active wave physics overcomes
the limits of passive classical systems.

2 Conclusion: The Trained Ear as Superior AI Detection

The preceding sections have shown that the auditory system is a physical processing system operating at the ξ^7 rung of T0 geometry — beyond the limits of passive mathematical analysis, beyond the Heisenberg uncertainty for linear systems. This leads to a practical conclusion that is rarely heard in the current debate about AI-generated media:

> We would be well served to train our physical capabilities rather than relying exclusively on technical solutions. A trained human ear will be able to detect AI-generated fakes for decades to come — as long as we train it.

2.1 Why Technical Detection is Structurally Disadvantaged

AI-generated audio and video fakes are produced with the same mathematical tools that digital detection systems are supposed to use to identify them: neural networks, Fourier analysis, statistical pattern recognition. This is a structural problem: the attacker and the defender operate on the same layer — the digital, mathematical, symbol-processing layer.

This creates an arms race that the attacker structurally wins: a new generator architecture renders the old detector worthless. The detector is updated. The generator is improved. The time lag between attack and detection grows with the complexity of the system.

> The human ear operates on a different layer: the physical, subatomic, ξ^7-encoded layer. AI generators produce signals in the digital domain. The physical signatures of real voices, real rooms, real instruments arise from mechanical processes at the atomic scale — and these signatures are not fully reproducible in the digital domain.

2.2 What a Trained Ear Detects

A vocal sound arises from vibration of the vocal folds, resonance of the vocal tract tube, radiation through lips and nose. Each of these processes is mechanical, non-linear, dependent on physiology and condition of the day. The acoustic signature of a real voice carries the fingerprints of this mechanics:

- Micro-fluctuations in fundamental frequency and amplitude (jitter, shimmer) — chaotic, non-reproducible, arising from airflow and muscle tension
- Room acoustic convolution: reflections, diffusion, absorption patterns of the physical space in which speech was produced
- Articulatory coarticulation: transitions between phonemes arise from inertia and elasticity of tongue, lips, palate
- Breath noise texture: turbulent flow in throat and mouth, unique every time

- Subharmonics and noise components from real vocal fold closure impulses

AI generators learn the statistical averages of these features from training data. They produce probable signals — but not physically generated ones. The absence of real mechanics leaves systematic traces: overly regular jitter distributions, missing room acoustic consistency, atypical coarticulation patterns, over-smoothed amplitude envelopes.

A layperson hears: “That sounds somehow odd.“ A trained ear hears: “The breath-pressure profile does not match the sentence rhythm.“ This is not a mystical talent — it is physical pattern recognition on the correct layer.

2.3 Training Instead of Trust in Technical Solutions

The verification of videos and audio recordings for AI fakes relies today predominantly on automated detection systems. This is problematic for the same reasons as digital audio analysis versus physical cochlear processing: the technical solution operates on the wrong layer.

Approach	Layer	Arms race?	Temporal advantage
Automated AI detector	Digital / mathematical	Yes — generator wins structurally	Months of lag
Trained human ear	Physical / ξ^7-encoded	No — different layer, not attackable	Immediate, stable for decades
Combined: ear + detector	Both layers	Partly — physical layer unchangeable	Optimal

Comparison of approaches to detecting AI-generated fakes.

What follows from this? Training. Not more technology, but more awareness of what the ear can do and how to sharpen it. Radio journalists and sound engineers develop this ear in their daily work. It is no secret how: active listening, conscious comparison of real and synthetic voices, analysis of room acoustics, work with unprocessed material.

2.4 The Trained Ear as Superior Detector — for Decades

AI generators will improve. But they will improve in the digital domain: they will statistically approximate real voices more closely. They will not improve at simulating real mechanics — because real mechanics is not digital. The vocal fold turbulence, the room impulse response, the breath-pressure profile: these arise in the physical world, at the ξ^7 scale, and no generator operating in the digital domain can fully reproduce them.

The trained ear detects this missing physics. Not because it computes better — but because it is itself physical. It is a ξ^7 system detecting ξ^7 signatures. No digital system can attack this level, because it operates on a different level.

Conclusion: We already possess the superior AI detector — it sits between our ears. It operates on the physical layer, is not attackable by digital countermeasures, works in real time without latency, and improves through training, not through computing power.

The question is not whether we should trust technology. The question is why we have stopped trusting our own ears.

The human ear is a physical quantum processor at the ξ^7 scale of T0 geometry. It operates beyond the Heisenberg uncertainty of passive systems, beyond the frame limits of digital sampling, beyond the reach of digital forgeries. Let us train it.